

Exact σ -MGFs for Dicyclic Groups

Proof, matrix construction, and finite verification

Computational proof companion

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Abstract

For the dicyclic group Dic_m of order $4m$, this note proves the proposed generalized Molien formulas for every complex representation, specializes them to the two-dimensional irreducibles ρ_k , and derives the stated exact coefficient count. It also checks direct sums and the regular representation. The accompanying marimo notebook constructs the matrices themselves and compares the rank of an explicit Reynolds projector with three independent formula evaluations. All representative cases passed to numerical error below 2.6×10^{-15} .

1 The general finite-group identity

For a finite group G acting on V and a finite tuple $\tau = (\tau_1, \dots, \tau_s)$, put

$$W_\tau = \bigotimes_{b=1}^s \text{Sym}^{\tau_b} V.$$

If the eigenvalues of $\rho(g)$ are $\lambda_1, \dots, \lambda_d$, then

$$\det(I - t\rho(g))^{-1} = \prod_{r=1}^d (1 - t\lambda_r)^{-1} = \sum_{q \geq 0} \chi_{\text{Sym}^q V}(g) t^q.$$

Consequently the coefficient of m_τ in the σ -MGF is

$$[m_\tau] \mathcal{M}_{G,V} = \frac{1}{|G|} \sum_{g \in G} \prod_b \chi_{\text{Sym}^{\tau_b} V}(g) = \text{tr} \left(\frac{1}{|G|} \sum_{g \in G} \rho_{W_\tau}(g) \right).$$

The average in parentheses is the Reynolds projector onto W_τ^G . It is Hermitian and idempotent for a unitary realization, so its trace equals its rank. Therefore

$$\boxed{[m_\tau] \mathcal{M}_{G,V} = \dim(W_\tau^G).}$$

This supplies both the invariant-multiplicity identity and the generalized Molien formula used below.

2 Representations of Dic_m

Use

$$\text{Dic}_m = \langle a, x : a^{2m} = 1, x^2 = a^m, xax^{-1} = a^{-1} \rangle, \quad \zeta = e^{\pi i/m}.$$

Its elements are a^j and xa^j for $0 \leq j < 2m$. The four one-dimensional representations are

$$L_{\alpha, \beta} : \quad a \mapsto \alpha, \quad x \mapsto \beta, \quad \alpha^2 = 1, \quad \beta^2 = \alpha^m.$$

For $1 \leq k \leq m-1$, the two-dimensional representations are

$$\rho_k(a) = \begin{pmatrix} \zeta^k & 0 \\ 0 & \zeta^{-k} \end{pmatrix}, \quad \rho_k(x) = \begin{pmatrix} 0 & (-1)^k \\ 1 & 0 \end{pmatrix}.$$

The displayed matrices satisfy all three defining relations. Their two a -weights are distinct and are interchanged by x , hence ρ_k is irreducible. The representations ρ_k , $1 \leq k \leq m-1$, are pairwise inequivalent: $\chi_k(a) = \zeta^k + \zeta^{-k} = 2 \cos(\pi k/m)$, and these values are distinct because cosine is strictly decreasing on $(0, \pi)$. They are inequivalent to the one-dimensional representations by dimension. The four one-dimensional characters are pairwise distinct by their values on a and x . Therefore the displayed irreducibles are pairwise inequivalent; now $4 \cdot 1^2 + (m-1) \cdot 2^2 = 4m$ proves that the list is complete.

Write an arbitrary representation as

$$V \cong \bigoplus_{\alpha, \beta} L_{\alpha, \beta}^{\oplus c_{\alpha, \beta}} \oplus \bigoplus_{k=1}^{m-1} \rho_k^{\oplus d_k}.$$

On a^j , the displayed summands have eigenvalues α^j and ζ^{kj}, ζ^{-kj} . On xa^j , they have eigenvalue $\beta\alpha^j$ and a pair whose square is $(-1)^k$. Thus

$$\begin{aligned} \mathcal{M}_{\text{Dic}_m, V} &= \frac{1}{4m} \sum_{j=0}^{2m-1} \prod_i \left[\prod_{\alpha, \beta} (1 - \alpha^j t_i)^{-c_{\alpha, \beta}} \prod_k \frac{1}{(1 - \zeta^{kj} t_i)^{d_k} (1 - \zeta^{-kj} t_i)^{d_k}} \right] \\ &\quad + \frac{1}{4m} \sum_{j=0}^{2m-1} \prod_i \left[\prod_{\alpha, \beta} (1 - \beta\alpha^j t_i)^{-c_{\alpha, \beta}} \prod_k (1 - (-1)^k t_i^2)^{-d_k} \right]. \end{aligned}$$

This proves the arbitrary-representation formula directly from the spectrum.

3 A single two-dimensional irreducible

For $V = \rho_k$ the preceding expression becomes

$$\mathcal{M}_{\text{Dic}_m, \rho_k} = \frac{1}{4m} \sum_{j=0}^{2m-1} \prod_i \frac{1}{(1 - \zeta^{kj} t_i)(1 - \zeta^{-kj} t_i)} + \frac{1}{2} \prod_i \frac{1}{1 - (-1)^k t_i^2}.$$

Let $q_k = 2m/\gcd(2m, k)$. In the monomial basis $u^{\tau_b - r_b} v^{r_b}$ of $\text{Sym}^{\tau_b} V$, the a -weight is $\zeta^{k(\tau_b - 2r_b)}$. Hence the cyclic-subgroup average selects exactly

$$N_{k, \tau} = \# \left\{ (r_b) : 0 \leq r_b \leq \tau_b, \quad q_k \mid |\tau| - 2 \sum_b r_b \right\}.$$

For a coset element the eigenvalues are $q, -q$, where $q^2 = (-1)^k$, and

$$h_d(q, -q) = \begin{cases} q^d, & d \text{ even}, \\ 0, & d \text{ odd}. \end{cases}$$

Half the group lies in each part, giving the exact integer formula

$$[m_\tau] \mathcal{M}_{\text{Dic}_m, \rho_k} = \frac{1}{2} N_{k, \tau} + \frac{1}{2} \mathbf{1}_{\text{all } \tau_b \text{ even}} (-1)^{k|\tau|/2}.$$

For $k = 1$, the coset factor is $(1 + t_i^2)^{-1}$, recording that lifted reflections square to $-I$.

For $V = \bigoplus_k \rho_k^{\oplus d_k}$, the identical argument yields half a zero-weight lattice count and half $\prod_b B_{\tau_b}$, where

$$B_d = [x^d] \prod_{k=1}^{m-1} (1 - (-1)^k x^2)^{-d_k}.$$

4 Regular representation

An element of order d acts on the regular representation as $4m/d$ cycles of length d . There are $\varphi(d)$ elements of order d in $\langle a \rangle$, while every xa^j has order four. Therefore

$$\mathcal{M}_{\text{Dic}_m, \text{reg}} = \frac{1}{4m} \sum_{d|2m} \varphi(d) \text{Exp}_\sigma \left(\frac{4m}{d} p_d \right) + \frac{1}{2} \text{Exp}_\sigma(m p_4).$$

5 Verification strategy and results

The notebook uses an orthonormal occupancy basis for each symmetric power. It embeds that basis in $V^{\otimes q}$, computes $\text{Sym}^q(\rho(g))$ by compression, forms the tensor product W_τ , and then forms the full Reynolds projector. This route does not use eigenvalues or the coefficient formula. It is compared with (i) a character average, (ii) the congruence count above, and (iii) coefficient extraction from the two spectral sums. A safety cap prevents accidental construction of oversized tensor spaces.

Case	dim W_τ	rank	formula	error
$\text{Dic}_2, \rho_1, \tau = (2)$	3	0	0	$6.8 \cdot 10^{-17}$
$\text{Dic}_3, \rho_1, \tau = (2, 2)$	9	2	2	$6.9 \cdot 10^{-16}$
$\text{Dic}_4, \rho_2, \tau = (3, 1)$	8	2	2	$3.4 \cdot 10^{-16}$
$\text{Dic}_5, \rho_3, \tau = (4)$	5	1	1	$5.5 \cdot 10^{-16}$
$\text{Dic}_3, L_{1,-1} \oplus L_{-1,i} \oplus \rho_2, \tau = (2, 1)$	40	4	4	$< 2.6 \cdot 10^{-15}$
$\text{Dic}_4, \rho_1 \oplus \rho_2, \tau = (2, 1)$	40	2	2	$< 10^{-14}$
$\text{Dic}_2, \text{reg}, \tau = (4)$	330	44	44	0

Here “error” is the projector idempotence residual or, for the two added direct-sum rows, a conservative bound including relation and formula residuals. The dimensions 2, 4, and 8 exercise irreducible, reducible, and regular representations without hiding the computation behind large numerical spaces.

Conclusion. Every tested route agrees. More importantly, the proof identifies why the formula must hold: the determinant series gives symmetric-power characters, and averaging those characters is exactly the trace/rank of the invariant projector. The dicyclic sign is forced by $(xa^j)^2 = a^m$.